

A Simple Algorithm for Jamming Detection in OFDM Systems

Krzysztof Wesołowski
Institute of Radiocommunications
Poznan University of Technology
Poznan, Poland
krzysztof.wesolowski@put.poznan.pl

Abstract—This paper presents a new and simple algorithm applied to detect jamming in OFDM transmission. A new metric is proposed for jamming detection. It can be used in both the time and frequency domains. It can also be implemented separately in each physical resource block, enabling the detection of passband jamming. Selected simulation results are presented showing the performance of the physical downlink shared channel in the MIMO 5G system configuration in the presence or absence of jamming.

Index Terms—Jamming, OFDM transmission, OFDM jamming, 5G systems

I. INTRODUCTION

Introducing the fifth generation (5G) wireless systems opened new areas of their applications and services. In general, three basic types of service can be distinguished in 5G systems: enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC). In some services, ensuring security is crucial to enabling them. An example of such a service is the application of 5G systems in wireless industrial URLLC control, vehicular-to-vehicular communications (V2V), or remote surgery. Among many possible attacks on the 5G networks that implement such services, jamming is one of the basic methods that can deteriorate or even destroy its functioning. In this paper, we will concentrate on the method of jamming detection in the context of a radio link of the 5G system. In contrast to many papers devoted to jamming in which incomplete systems are considered, we will instead investigate the influence of jamming and its detection on the overall performance of the main data transmission channel in 5G links, namely the physical downlink shared channel (PDSCH).

There are several different attack strategies that can be performed by a jamming device that creates interference to the regular link. The newest survey of jamming attacks and antijamming strategies in wireless networks can be found in [1]. Another survey can be found in [2]. Smart jamming attacks on the 5G New Radio are the subject of [3]. The existing jamming attack strategies can be roughly categorized into the following kinds [4]:

- **Constant jammer** continuously emits a radio signal that represents random bits that do not follow any MAC protocol.
- **Deceptive jammer** transmits semi-valid packets consisting of a valid header and random payload.
- **Random jammer** alternates between sleeping and jamming the channel. In the sleeping mode, the jammer does not

generate any distorting signal. In the jamming mode, it operates as a constant or deceptive jammer.

- **Reactive jammer** tries to minimize the energy of the jamming signal by detecting useful transmission and then jamming it. It insists on modifying enough bits of the user packet to make it classified by the receiver as a non-valid one.

In this paper, we investigate the detection of **random jamming** in the form of a noise signal or an OFDM signal similar to that transmitted by the downlink of the 5G system. To minimize the influence of jamming on overall link performance, the main task is to reliably detect that jamming occurs.

Several jamming detection techniques have been reported in the literature. They can be categorized into two main classes [5]: non-machine learning and machine learning. The interesting list of papers representing both classes can be found in [5]. In non-machine learning methods, some parameters and strategies are applied, such as thresholds [6], fuzzy logic [7], game theory, channel surfing, mapping of the jammed region [7], and timing channel [8].

Machine learning strategies for jamming detection have been reported in several papers (see [9], [10], [11], and [5] as examples). In our paper, we propose a jamming detection algorithm that belongs to non-machine learning methods.

II. SYSTEM MODEL

Our investigations were performed on the example of a typical downlink for the 5G system. We considered the whole chain of functional blocks for the physical downlink shared channel. The MIMO OFDM technology was applied with N_T antennas at the base station and N_R antennas at the terminal. LDPC coding that is consistent with the 5G NR standards was also used. Fig. 1

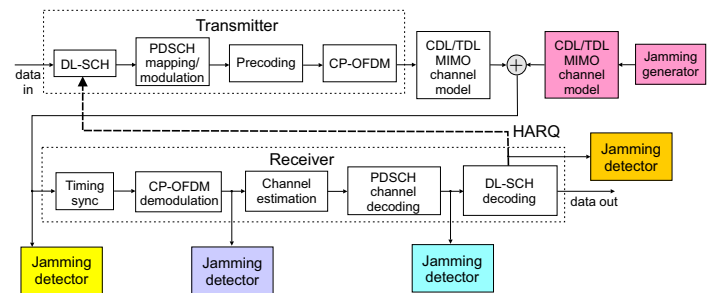


Fig. 1. The model of the PDSCH transmission in a 5G system

This work was funded by National Center for Research & Development under Grant No. CYBERSECIDENT/487845/IV/NCBR/2021.

shows the block diagram of the system that was the subject of our experiments.

We note that the jamming signal is injected into the transmission chain as the additive signal from the output of the jamming MIMO channel that transmits the signal from the jamming signal source (see the magenta blocks in Fig. 1).

Let us note the potential locations for jamming detection. The first marked in yellow shows the possibility of jamming detection in the time domain, still before demodulation. Operations performed in this place enable potential detection of the existence of jamming; however, it is not possible to differentiate if jamming has a wideband or narrowband nature and, in its last case, where it is located on the frequency axis. The second possible location of the jamming detector shown as a violet box is on the output of the OFDM demodulator. The advantage of placing the jamming detector at this location is the ability to identify physical resource blocks (PRBs) that have been infected by jamming. However, the price to pay for this method is to perform jamming detection in parallel for all PRBs or at least for their blocks assigned to specific users.

Finally, the other two locations use the indicators associated with channel coding. Abnormal frequency of wrong channel decoding compared with the previous situations or more frequent requests for incremental redundancy block repetition indicate the appearance of jamming, too. We note that a jamming detection method can also be applied that uses some combined jamming indicators jointly. In this paper, we concentrate on the simple detectors applied in the time and frequency domains.

III. PROPOSED JAMMING DETECTION METHOD

Let us analyze the OFDM signal received in two neighboring time intervals of the appropriately selected time length. An example of such an interval can be an OFDM slot, which in the case of 5G NR consists of 14 OFDM symbols. The two subsequent intervals of analysis can slide in time or be synchronized with the received OFDM signal. For convenience in our simulation experiment, we assume the second case.

The simplest method of jamming detection can be described in the following way. Typically, when there is no jamming, the mean power (or energy) in two subsequent intervals remains approximately constant. The appearance of jamming causes an increase in the measured energy. It can be detected by setting the appropriate threshold and checking if the energy of the received signal crosses it or not. Unfortunately, such a threshold has to be continuously adjusted to the current signal-to-noise ratio or scaled with respect to the received signal power and can be different for several distances of the terminal from the base station. Instead, we propose applying the detection criterion, which can be described by the following formula.

$$d_n = \ln \frac{E_n}{E_{n-1}} \quad (1)$$

where E_n and E_{n-1} are the energies of the signal samples received and collected at the n -th and $(n-1)$ -st intervals on all the receive antennas, respectively. Thus,

$$E_n = \sum_{k=1}^{N_R} \sum_{i=1}^m |r_{k,(n-1)m+i}|^2 \quad (2)$$

where k is the index of the receive antenna and i is the current number of the signal sample received in the n -th interval. We assume that the received sample is given by the following formula:

$$r_{k,j} = \begin{cases} x_{k,j} + \nu_{k,i} + \zeta_{k,j} & \text{if jamming is present} \\ x_{k,j} + \nu_{k,j} & \text{if jamming is absent} \end{cases} \quad (3)$$

and $j = (n-1)m + i$. The symbols $x_{k,j}$ denote the samples of the useful received signal, $\nu_{k,j}$ are the additive Gaussian noise samples, and $\zeta_{k,j}$ are the jamming signal samples. Note that if jamming occurs in two analyzed n -th and $(n-1)$ -st intervals or not, the measure d_n in (1) is around zero assuming that the desired signal, jamming and noise statistics remain constant. However, if jamming appears in the n -th interval and has not appeared in the previous one, the measure d_n is positive, while in the opposite situation, d_n is negative. The crucial task is to find the appropriate thresholds between particular ranges of the metric d_n assigned to detection decisions. Denote the thresholds by $\pm\Delta$. Thus, the detection decision D_n can be based on the following rule:

$$D_n = \begin{cases} \text{if } d_n < -\Delta & \text{then } D_n = 0 \\ \text{if } d_n > \Delta & \text{then } D_n = 1 \\ \text{if } -\Delta \geq d_n \geq \Delta & \text{then } D_n = D_{n-1} \end{cases} \quad (4)$$

where $D_n = 1$ denotes the detection of jamming in the n -th time interval, and $D_n = 0$ indicates the decision that the interval is free from jamming.

A precise analysis of the proposed metric requires the derivation of its probability density function, which depends on the statistical nature of the jamming. Basically, two types of jamming signal can be considered. The first type is an additive noise, while the second is another OFDM signal. The jamming signal is sent over the channel between the jammer and the receiver. The probability density function is useful for finding the appropriate threshold values $\pm\Delta$. Note that the thresholds do not depend on the signal-to-noise ratio to that extent as is in a traditional approach to threshold jamming detection. We simply compare the logarithmic ratio of the measured energies in two subsequent time intervals.

Recall again the energy signal applied in (1). It can be expressed in the following form.

$$\begin{aligned} E_n &= \sum_{k=1}^{N_R} \sum_{j=(n-1)m+1}^{nm} |r_{k,j}|^2 = \sum_{k=1}^{N_R} \sum_{j=(n-1)m+1}^{nm} |x_{k,j} + \nu_{k,j} + \zeta_{k,j}|^2 \\ &= \sum_{k=1}^{N_R} \sum_{j=(n-1)m+1}^{nm} |x_{k,j}|^2 + \sum_{k=1}^{N_R} \sum_{j=(n-1)m+1}^{nm} |\nu_{k,j}|^2 \\ &\quad + \sum_{k=1}^{N_R} \sum_{j=(n-1)m+1}^{nm} |\zeta_{k,j}|^2 + \text{cross-terms} \end{aligned} \quad (5)$$

Based on the central limit theorem, we can assume that the samples of the OFDM signal $x_{k,j}$ are approximately complex Gaussian random variables. Thus, the first term of the double sum in (5) is the sum of $N_R m$ Rayleigh-distributed random variables. Similarly, we can say the same about the noise samples and jamming samples if jamming is currently applied. There is no analytical formula for the probability distribution of the sum of Rayleigh-distributed random variables. A good analytical approximation to this distribution can be found in [12] (see formula (9) in it). The remaining cross-terms can be considered

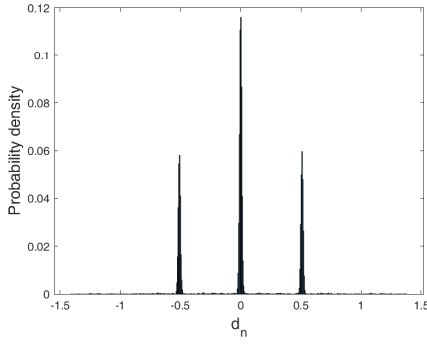


Fig. 2. Probability density function of the metric d_n for SNR= 3dB and jamming level equal 6 dB above additive noise level

as approximately Gaussian-distributed. In general, the probability distribution of the sum of random variables is the convolution of the probability distributions of all components, so the probability distribution of E_n or E_{n-1} is analytically intractable. The final probability distribution of $d_n = \ln(E_n/E_{n-1})$ is a function $g(E_n, E_{n-1})$ of the distributions of the measured energies in the intervals n -th and $n-1$ -st and is also not manageable. Therefore, in deriving the thresholds $\pm\Delta$, we resort to simulations. In Figs. 2 and 3 we show the estimated probability densities of the metric d_n derived in the simulated link when the received signal is corrupted by an OFDM jamming signal. The jamming signal performed is a type of random jamming in which time slots are the subject of jamming with a probability equal to 0.5. The SNR measured at receiver input (in the time domain) in both cases was equal to 3 dB, while in Fig. 2 the jamming level was 6 dB higher than the noise level, while in Fig. 3 it was at the noise level. Similar plots can be obtained for other values of SNR. On the basis of them, we have selected the decision thresholds $\pm\Delta = \pm 0.05$.

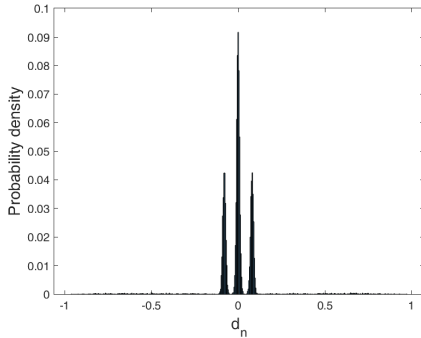


Fig. 3. Probability density function of the metric d_n for SNR= 3dB and jamming level equal to additive noise level

The factor that decides on the selection of thresholds is, rather, the distribution of d_n around zero, which depends on the number of summed samples contained in the intervals analyzed. Changing the level of the jamming signal causes the density sidelobes to shift with respect to the peak located around zero. As we can see, the decision regions are clearly differentiable, unless the level of the jamming signal is small enough to the extent that jamming and noise are visualized in a single peak. However, in such a

situation, we expect that jamming detection is not needed, as it will not cause deterioration of the PDSCH channel throughput due to the powerful channel coding applied.

The probability distributions shown in Figs. 2 and 3 have been derived in the time domain. Similar results can be obtained if the metric is calculated on the basis of the samples at the FFT demodulator output. As we have already mentioned, such a method is more flexible, as we can detect jamming in particular PRBs contrary to the time domain calculations where only the existence of jamming can be detected. However, because the number of samples processed in the metric calculation for particular PRBs is smaller compared to the time-domain calculations, the thresholds have to be adjusted accordingly.

IV. MODEL OF THE SIMULATED SYSTEM

The concept of the proposed jamming detector was verified on the basis of the 5G PDSCH system model offered in the Matlab 5G toolbox that was appropriately modified for our investigations. The model conforms to the basic 3GPP specifications.

Due to space limitations, we concentrate on the selected system configuration features only. We applied a 20 MHz transmission model with 30 kHz space between subcarriers. Thus, 51 PRBs with 12 subcarriers each are used in the downlink. In the transport downlink shared channel (DL-SCH) a standard LDPC coding is applied. Each subcarrier is QPSK modulated. As a result of the modulation and coding scheme selection, the full throughput is 30.224 Mbps. The base station uses $N_T = 8$ antennas, while the terminal operates with $N_R = 2$ antennas. The precoding based on SVD calculations is applied jointly for all PRBs. Sixteen HARQ processes are applied in the data transmission strategy. The applied channel model is based on the clustered delay line (CDL) principle (see [13]). The CDL-C model (urban macrocell model) is applied with a delay spread equal to 300 ns. In the simulations, the ideal channel estimation was assumed; thus, the achieved throughput values are rather upper limits of the real ones.

Although both jamming sources, namely noise and OFDM signals, were investigated, in this paper, we concentrate on the detection of OFDM jamming signals only. All OFDM subcarriers were fully QPSK randomly modulated. Two possible characteristics of the OFDM signal jamming were investigated. In the first case, we assumed that the signal covers the entire spectrum of the useful signal, that is, all 51 PRBs are subject to jamming. In the second case, we considered a narrowband jamming signal that corrupted only one PRB, and its entire power was concentrated in a 360 kHz band of a single PRB. In this case, to identify the jamming location, detection was performed in parallel for each PRB, so 51 concurrent detection processes were applied. The OFDM jamming signal was transmitted through the jamming channel represented by the 2x2 MIMO model of the tapped delay line (TDL). TDL-A profile was selected and the delay spread was equal to 100 ns.

V. SIMULATION RESULTS

As mentioned above, both time- and frequency-domain detection processes were performed. In the following, we show selected plots of the throughput achieved and probabilities of jamming detection for several values of SNR and different values of jamming levels. Jamming is generated on the slot basis, and

the probability of its appearance in a slot is assumed to be equal to 0.5; therefore, the lowest throughput bound is half of the value resulting from the current SNR value and system tolerance to jamming. In all plots, throughput is given as a percentage of its maximum value resulting from the applied modulation and coding scheme. All plots are drawn as functions of DSNR, that is, the level of the jamming signal above the SNR. The following dependence between SNR the signal-to-interference (SIR) and DSNR can be defined:

$$SIR_{dB} = SNR_{dB} - DSNR_{dB} \quad (6)$$

All the following figures contain the plots of: throughput given as a percentage of its maximum value, probability of correct jamming detection, probability of false alarm (the algorithm detected jamming, whereas in fact it was absent), and probability of undetected jamming.

First, let us report the results when the wideband OFDM jamming signal is applied. Examples of the performance of the proposed detection algorithm are shown in Figs. 4 and 5. The algorithm operates in the time domain. Similar simulation results

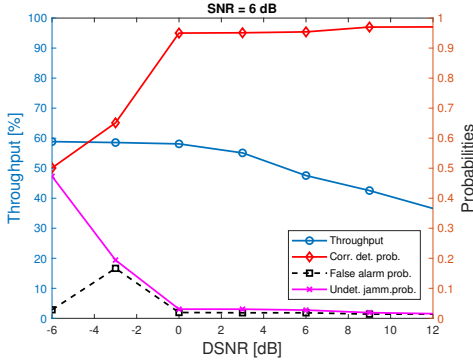


Fig. 4. Throughput percentage, probability of correct jamming detection, false alarm, and undetected jamming for SNR = 6 dB versus DSNR

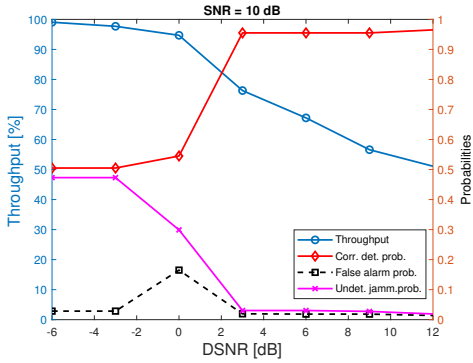


Fig. 5. Throughput percentage, probability of correct jamming detection, false alarm, and undetected jamming for SNR = 10 dB versus DSNR

were received for narrowband jamming OFDM signal detection using the proposed algorithm. The 10th PRB was the subject of random jamming with a probability of 0.5 in a series of slots. Figs. 6 - 11 show the throughput and the set of the probabilities mentioned above for a few values of SNR (equal 0, 6, and 10 dB)

versus DSNR when the detection operation is performed in the time and frequency domains, respectively. Note that due to the different ranges of summed squares of values in the output of the FFT demodulator used by the detection algorithm, the thresholds $\pm\Delta$ had to be adjusted appropriately. The algorithm (1) was found to work very well when the threshold had value $\Delta = 1.5$.

One can easily note that the proposed algorithm is very reliable in the relevant range of jamming levels. It does not detect jamming if its level is very low (the probability of its correct detection is then about 0.5); however, the throughput in PDSCH has not deteriorated then. When the level of the jamming signal starts to increase, the proposed algorithm working according to (1) detects the existence of jamming very reliably. The probability of correct detection is at least 0.95 or greater. We also note that the operation per PRB in the frequency domain is very reliable. The jamming detection quality achieved is even higher than that in the time domain. Additionally, as we have already mentioned, we gain the possibility of identifying jamming in particular PRBs and finding jamming-free PRBs. This can be useful as a countermeasure against jamming because the system control unit can order hopping to another PRB which has not been attacked by the narrowband jammer.

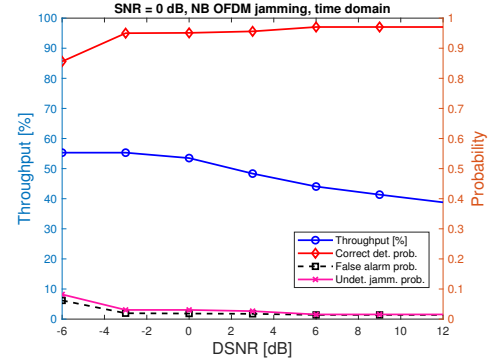


Fig. 6. Throughput percentage, probability of correct jamming detection in time domain, false alarm, and undetected jamming for SNR = 0 dB versus DSNR

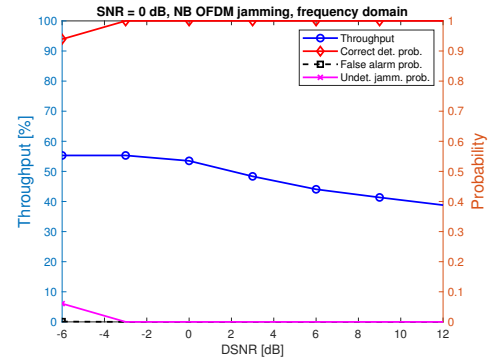


Fig. 7. Throughput percentage, probability of correct jamming detection in frequency domain, false alarm, and undetected jamming for SNR = 10 dB versus DSNR

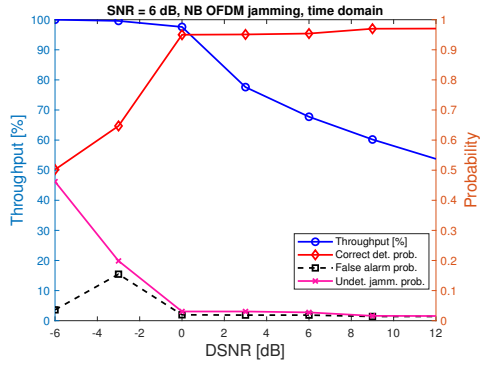


Fig. 8. Throughput percentage, probability of correct jamming detection in time domain, false alarm, and undetected jamming for SNR = 0 dB versus DSNR

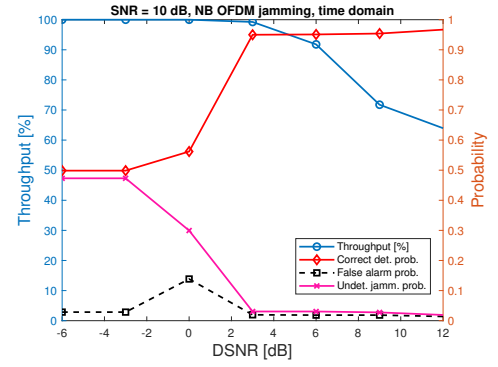


Fig. 10. Throughput percentage, probability of correct jamming detection in time domain, false alarm, and undetected jamming for SNR = 10 dB versus DSNR

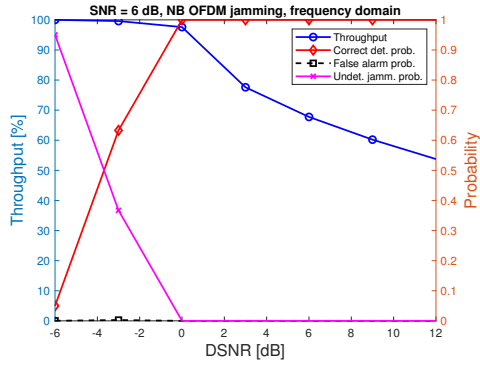


Fig. 9. Throughput percentage, probability of correct jamming detection in frequency domain, false alarm, and undetected jamming for SNR = 6 dB versus DSNR

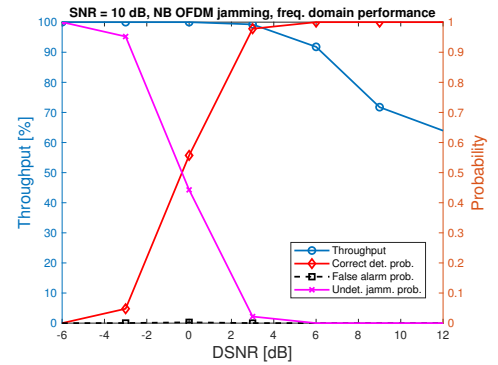


Fig. 11. Throughput percentage, probability of correct jamming detection in frequency domain, false alarm, and undetected jamming for SNR = 10 dB versus DSNR

VI. CONCLUSIONS

In this paper, we have proposed a very simple and effective jamming detection algorithm that can be useful in a multi-tone transmission. Simulation experiments have been performed on the example of a 5G system downlink; however, this method is universal and can be applied in other systems based on OFDM transmission. Additionally, the proposed method can work as a support of jamming detection methods operating in higher layers (e.g. using the abnormal behavior of the HARQ processes, or received packet analysis).

REFERENCES

- [1] H. Pirayesh, and H. Zeng, "Jamming Attacks and Anti-Jamming Strategies in Wireless Networks: A Comprehensive Survey," *IEEE Comm. Surv. Tutorials*, vol. 24 No.2, 2022, pp. 767-809
- [2] K. Grover, A. Lim, and Q. Yang, "Jamming and Anti-Jamming Techniques in Wireless Networks: a Survey," *Int. J. of Ad Hoc and Ubiquitous Computing*, Vol. 17, No. 4, Dec. 2014, pp. 197-215
- [3] Y. Arjoune, and S. Faruque, "Smart Jamming Attacks in 5G New Radio: A Review," in Proc. 2020 10th Ann. Computing and Commun. Workshop and Conf. (CCWC), 6-8 Jan. 2020, Las Vegas, USA
- [4] A. Hamieh, and J. Ben-Othman, "Detection and Jamming Attacks in Wireless Ad Hoc Networks Using Error Distribution", in Proc. 2009 IEEE Int. Conf. Commun. (ICC), 14-18 June 2009, Dresden, Germany pp.1-6
- [5] Y. Arjoune, F. Salahdine, Md. Sh. Islam, E. Ghribi, and N. Kaabouch, "A Novel Jamming Attacks Detection Approach Based on Machine Learning for Wireless Communication", in Proc. 2020 Int. Conf. on Information Networking (ICOIN), 7-10 Jan. 2020, Barcelona, Spain, pp.1-6

- [6] M. Cheng, Y. Ling, and W. Wu, "Time Series Analysis for Jamming Attack Detection in Wireless Networks", in Proc. the 2017 IEEE Global Comm. Conf. (GLOBECOM), 4-8 Dec. 2017, Singapore
- [7] S. Khattab, D. Mosse, and R. Melhem, "Modeling of the Channel-Hopping Antijamming Defense in Multi-Radio Wireless Networks", in Proc. 5th Ann. Int. Conf. on Mobile and Ubiquitous Systems: Computing, Networking, and Services, MobiQuitous 2008, July 21-25, 2008, Dublin, Ireland, pp.1-6
- [8] H. Reyes, and N. Kaabouch, "Jamming and Lost Link Detection in Wireless Networks with Fuzzy Logic," *Int. J. of Scientific Eng. Research*, vol. 4, pp. 1-7, 2013
- [9] T. Nawaz, D. Campo, M. O. Mughal, L. Marcenaro, and C. S. Regazzoni, "Jammer Detection Algorithm for Wide-band Radios Using Spectral Correlation and Neural Networks," in Proc. 2017 13th Int. Wireless Comm. and Mobile Computing Conf. (IWCMC), 26-30 June 2017, Vanelcia, Spain, pp. 246-251
- [10] Y. Arjoune, and S. Faruque, "Real-time Machine Learning Based on Hoeffding Decision Tree for Jamming Detection in 5G New Radio," in Proc. 2020 IEEE Int. Conf. on Big Data, 10-13 Dec. 2020, Atlanta, USA
- [11] M. Hachimi, G. Kaddoum, Gh. Gagnon, and P. Illy, "Multi-stage Jamming Attacks Detection Using Deep Learning Combined with Kernelized Support Vector Machine in 5G Cloud Radio Access Networks," in Proc. 2020 Int. Symp. on Networks, Computers and Commun. (ISNCC), 20-22 Oct. 2020, Montreal, Canada, pp. 1-5
- [12] J. Hu, and N. Beaulieu, "Accurate Simple Close-Form Approximations to Rayleigh Sum Distributions and Densities," *IEEE Comm. Letters*, vol. 9, No. 2, Febr. 2005, pp. 109-111
- [13] 3GPP TR 38.901. "Study on channel model for frequencies from 0.5 to 100 GHz (Release 15)." 3rd Generation Partnership Project; Technical Specification Group Radio Access Network.